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Review article

Review of research into bioregenerative life support system(s) which can support humans living in space

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ABSTRACT

To travel beyond the Earth and realize long-term survival in deep space, humans need to construct Bioregenerative Life Support System (BLSS), which reduces the requirement for supplies from the Earth by *in situ* regenerating oxygen, water and food needed by astronauts, and prevents pollution to extraterrestrial bodies by recycling waste. Since the 1960s, the USSR/Russia, the United States, Europe, Japan, and China carried out a number of studies with abundant achievements in BLSS systematic theories, plants/animals/microorganisms unit technologies, design/construction, and long-term operation/regulation. China's "Lunar Palace 365″ experiment realized Earth-based closed human survival for a year, with a material closure of >98%. However, a lot of research work is still needed to ultimately realize BLSS application in space, especially given the space experiment of BLSS never carried out, and the overall impact of space environment on BLSS unknown. Lunar exploration projects such as lunar village and lunar research station are successively proceeding. Therefore, future BLSS research will focus on lunar probe payload carrying experiments to study mechanisms of small uncrewed closed ecosystem in space and clarify the impact of space environmental conditions on the ecosystem, so as to correct the design and operation parameters of Earth-based BLSS. Such research will provide theoretical and technological support for BLSS application in crewed deep space exploration.

1. Introduction

"Earth is the cradle of humanity, but one cannot live in a cradle forever (Tsiolkovsky, 1912)." Various countries have released plans of crewed deep-space exploration such as lunar research stations and landing on Mars. To travel out of the Earth, go into deep space and achieve long-term survival, a Bioregenerative Life Support System (BLSS) must be built to minimize the need of supplies from the Earth by *in situ* circulating oxygen, water and food for astronauts, and to prevent pollutions to extraterrestrial bodies by recycling waste. BLSS is an artificial closed ecosystem composed of humans, plants, animals and microorganisms based on ecological principles, and also a combination of technologies including biotechnology and engineering control technology. BLSS has the same structure of producer (plants), consumer (humans/animals) and decomposer (microorganisms) as the ecosystem on the Earth's surface, which is able to recycle and regenerate oxygen, water, food and other essential substances needed by human survival, and provide humans with comfortable environments similar to that of the Earth's ecosystem (Fig. 1). BLSS is one of the key technologies for crewed deep space exploration, supporting long-term autonomous existence of humans (Liu et al., 2009).

2. BLSS's theoretical and technological system framework and technological research development path

Theories and technologies of BLSS are multidisciplinary, including biology, environmental engineering, ecology, nutrition, psychology, computer science, electronic information engineering, and materials science, etc. The research and development of BLSS theoretical and technological systems should be based on the knowledge system and research results of multidisciplinary fields, and guided by actual design, construction and operation requirements of BLSS. The framework of theoretical and technological system of BLSS is shown in Fig. 2.

The development path of technological research BLSS is shown in

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Fig. 3. Theories and technologies of individual units of BLSS will be studied first, and then theories and technologies of artificial ecosystem will be studied. On this basis, design and construction methods of the system will be established, and then the stage of experimental ecosystem research with human participation will be entered. The construction of Earth-based experimental BLSS and the experimental study of artificial closed ecosystem with humans as a biological unit are of important milestone, which marks the ability to support the survival material needs of humans through biological recycling and regeneration on the Earth. However, to apply Earth-based technologies to space, extraterrestrial experiments must be conducted to correct the design and operational parameters of Earth-based BLSS. In this way, BLSS technologies for extraterrestrial application can be established.

3. Research progress of earth-based BLSS

BLSS technologies for outer space exploration, since the 1960s, the Soviet Union/Russia, the United States, Europe, Japan, China and other countries/regions successively started research of bioregenerative life support technologies, set up BLSS experimental systems of different levels and types, and carried out Earth-based simulation experiments.

3.1. Earth-based BLSS research: the soviet union/russia

The research of BLSS was initiated by Institute of Biomedical Problems (IBMP), which built a crewed closed experiment with a 5 m^3 chamber containing three unicellular algae cultivators (15 L in volume each) as oxygen producer. The experiment proved that 90% of air could be regenerated by algae (Shepelev, 1966). In the 1960s, Institute of Biophysics (IBP) of the Siberian Branch of the Soviet Academy of Science built the BIOS system — the world's first large-scale Earth-based comprehensive experimental facility for the study of BLSS. The system initially consisted of only a 12 m^3 chamber, which was equipped with an 18 L microalgae culture reactor for gas circulation and regeneration. The major research purpose of this system was gas circulation, with all water and food needed provided by the outside (stored in the system in advance). Some experiments have shown that the system achieved partial closure in terms of gas circulation, and can meet about 20% of a person's material needs. To further improve the closure, the researchers expanded the original system by adding a 2.5 × 2.0 × 1.7 m plant chamber, thus upgrading the system to BIOS-2. The plant chamber provided some food while providing gas purification. After four years of system adjustment and a 90-day closed experiment in between, BIOS-2 demonstrated that the gas cycle can be fully closed in the system, with about 25% of O_2 regenerated by plants and the rest by microalgae. The water in the system was also cycled, with an overall closure of 80% to 85% (Salisbury et al., 1997).

In 1972, on the basis of the above work, researchers built a system with a volume of 315 m^3 named BIOS-3, in an attempt to achieve complete closure of the material cycle (Sychev et al., 2011). The BIOS-3 consisted of four compartments. The crew occupied a compartment subdivided into work area and three cabins: a kitchen, a toilet, and a control room. The other three compartments was used to grow plants. The BIOS-3 carried out crewed experiments for 4–6 months in winters of 1972, 1976 and 1983. The results showed that the 63 m^2 planting area enabled the system for $a > 90\%$ closure in terms of gas and water circulation, and met 40–45% of food need of three people (Gitelson et al., 1989) or 70% of two people (Gribovskaya et al., 1997). Moreover, these crewed experiments also realized the automatic adjustment of plant photosynthetic productivity in BIOS-3 following the system's interior environmental changes (Gitelson et al., 1976). BIOS-3 achieved 95% closure in some experiments, and this record held for >30 years (Gitelson et al., 1976; Salisbury et al., 1997). This work provided valuable reference for other countries to carry out relevant researches later.

Although the closed crewed experiments of BIOS ended in the 1980s,

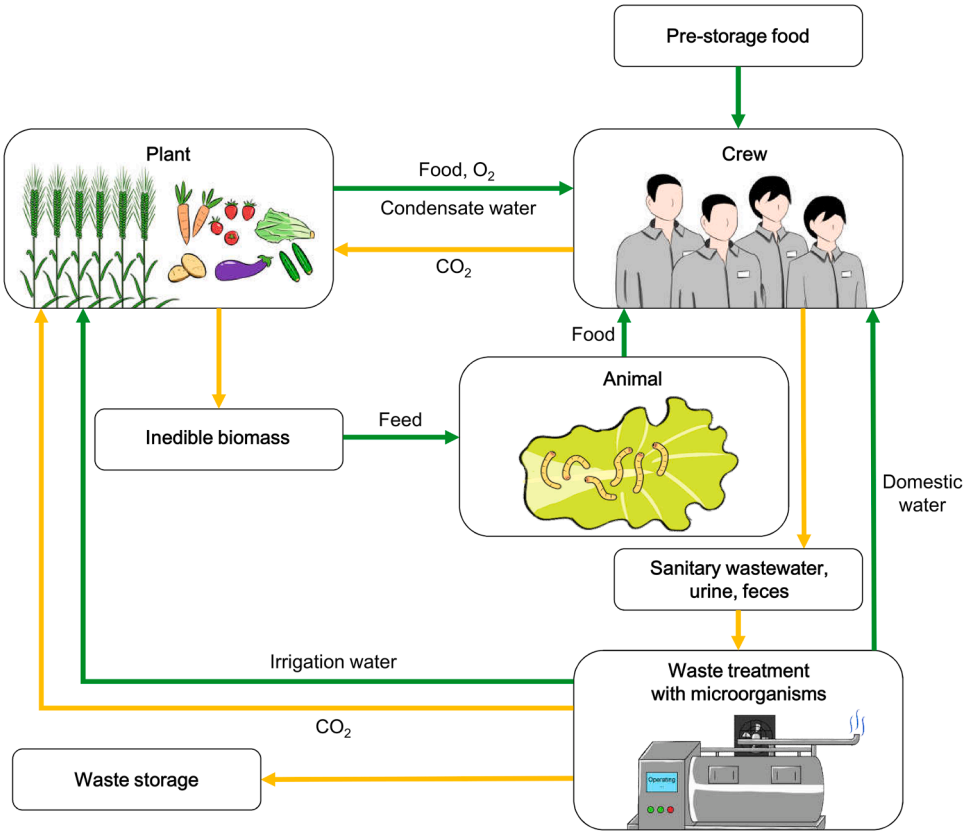

Fig. 1. Material cycles in BLSS.

they have demonstrated the feasibility of building a closed ecosystem based on bioregenerative life support technologies. It is the long-term pursuit of Russian scientists engaged in life support research to improve the efficiency of BLSS biological units, improve the degree of coupling between biological units, perfect the regulation strategy of biological units, and further improve the closure and stability of BLSS (Degermendzhi and Tikhomirov, 2014).

3.2. Earth-based BLSS research: the United states

In the early 1960s, the study of BLSS also aroused great interest among American scholars. In 1966, the National Aeronautics and Space Administration (NASA) began work on BLSS at the Ames Research Center, holding the first conference on conducting closed ecosystem research (Drake et al., 1966).

In 1979, NASA launched the Controlled Ecological Life Support System Program (CELSS), committed to life support research using ecological and biological theories and technologies comprehensively (Mitchell, 1994; Wallace and Powers, 1990). Several years later, NASA developed the Environmental Control and Life Support System (ECLSS) based on CELSS, and put forward the theory of Advanced Life Support (ALS) for missions such as lunar and Martian bases (Henninger et al., 1996; Paul and George, 1991).

In addition, collaborating with a number of American universities, NASA conducted plant cultivation experiments in closed environments, built a number of BLSS experimental facilities including the Biomass Production Module (BPC), built a system design database and thereby calculated the volume, equipment weight and energy consumption required to higher plant cultivation in BLSS (Drake et al., 1966; Wheeler et al., 2008, 1996). They concluded that 43 m^2 of higher plant cultivation is normally able to provide 90% of a person's nutritional requirement (Wheeler, 1997), which was planned as the design basis of Bio-Regenerative Planetary Life Support Systems Test Complex (BIO--Plex), but BIO-Plex was not built eventually.

From 1995 to 1997, NASA carried out the Lunar-Mars Life Support Test Project (LMLSTP), which integrated biological regenerative life support technologies with traditional physical/chemical technologies (Kloeris et al., 1998), including three stages of experiments. Among them, the first stage achieved one-person closed survival for 15 days, the second stage achieved four-person closed survival for 30 days and 60 days successively, and the third stage achieved four-person closed survival for 91 days. Wheat in the system provided 25% of oxygen the crew needed for breathing, and wheat and lettuce met 5% of the crew's food requirement (Barta et al., 2006). The results showed that higher plants and physical/chemical technologies had high integration in gas (air) regeneration. Water treatment efficiency of the bioreactor system reached the astronaut's drinking water standard (Henninger, 2002). In addition, LMLSTP carried out studies on psychological changes of the crew (Holland and Curtis, 1998).

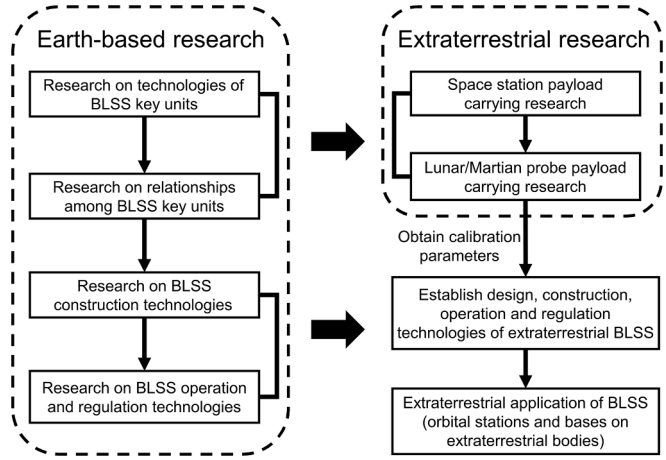

Fig. 3. BLSS's technological research development path.

Since the new century, NASA has started the Habitat Demonstration Unit - Deep Space Habitat (HDU-DSH) project, which established a habitat system in a desert of Arizona, USA. A series of crewed system experiments were carried out in 3 successive years (Kennedy, 2011), and included a supplemental food production system using plants (Massa et al., 2013). In related studies, NASA sponsored the development of a prototype, deployable Lunar Greenhouse (LGH) for possible use on future crewed missions to the moon or Mars (Boscheri et al., 2012).

In addition to NASA, multinational scientists also constructed Biosphere 2 in the desert of Arizona, USA from 1987 to 1991. Given that "Biosphere 1″ is the Earth's biosphere, Biosphere 2 included various ecosystem types of the Earth's biosphere such as rainforest, tropical grassland, desert, mangrove marsh and coral reef ocean, serving as a laboratory for global ecological processes, and also an early attempt of the kind of complex ecological and life support systems for long-term habitation in space (Nelson, 2018). Biosphere 2 is known for its first 24-month closure experiment September 1991–1993 and the second 6.5-month closure experiment March–September 1993, with eight

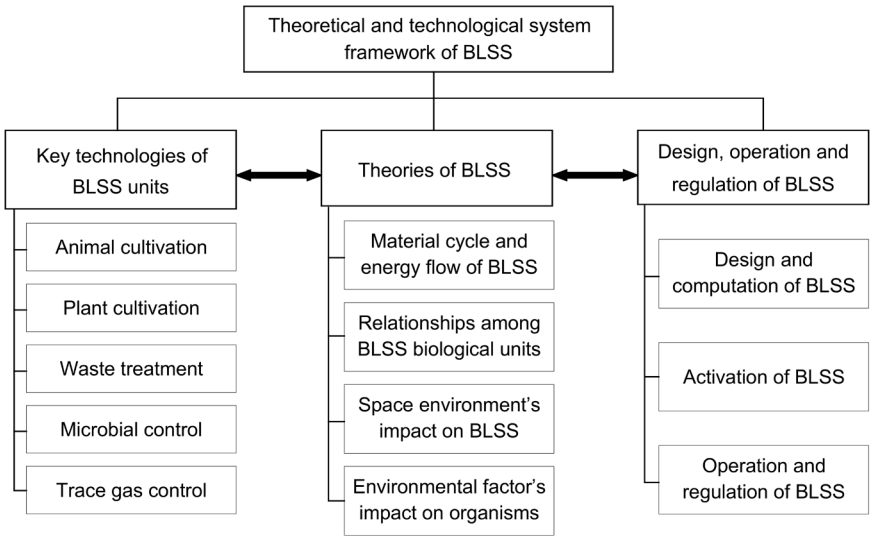

Fig. 2. Theoretical and technological system framework of BLSS.

people living in it (Nelson, 2021). The facility covers an area of 1.3 hectares, with a total volume of 1.8×10^5 m^3 (Dempster, 1999), and is the largest closed ecological system facility yet created (Nelson, 2021). Based on the experience of the first experiment, system-level balance of essential substances (air, water, and food) of the second experiment comprehensively improved. The first experiment produced 83% of the food needed, based on the experience and selection of better adapted crops, the second experiment produced 100% of the food needed (Marino et al., 1999). Unexpected carbon-dioxide absorption into unsealed concrete inside the facility happened during the first experiment (Nelson, 2018). Possibly resulted from the early imbalance of photosynthesis and respiration, atmospheric oxygen level slowly and continuously declined during the first 16 months of the first experiment, and was balanced by the injection of oxygen from outside; the injection of oxygen was no longer needed during the second experiment probably because plant biomass had greatly increased, and soils had also been matured and stabilized (Allen, 1997; Nelson et al., 1994). Exploring similarities and differences between the Earth's biosphere and artificial ecosystem design, solving various unexpected setbacks, Biosphere 2 contributed precious experience to BLSS development and the types of facilities which will be required for humanity to truly become multi-planetary with its positive history of exploration, discovery and challenge.

3.3. Earth-based BLSS research: Japan

In 1988, the Institute for Environmental Sciences of Japan started to plan for Closed Ecology Experiment Facilities (CEEF). The construction of the system began in 1994 in Rokkasho Village of Aomori Prefecture and completed in 1998. An important feature of CEEF was the introduction of the first animal unit, the goat. In terms of structure, CEEF is mainly composed of three parts: a closed plantation module, a closed animal breeding and habitation module, and a closed geo-hydrosphere material circulation experimental module. Each module can operate independently or in combination to carry out research. There was only energy and information exchange between the system and external environment (Masuda et al., 2005). In addition, it completely used physical and chemical equipment to treat waste, and material exchange among units was also carried out by physical and chemical equipment (Nitta, 2001). Computer technologies were used to analyze material circulation law (Miyajima, 2017). The advantage of its physical and chemical methods was the rapid circulation of substances in the system which facilitated the control, and the disadvantage was the ignorance of the important role of microorganisms. Moreover, the system was so large that it relied on external control.

From 2005 to 2007, CEEF successfully carried out a number of "human–goat–higher plants" short-term closed experiments (Tako et al., 2010), with the longest continuous experiment duration of four weeks. There have been no reports of subsequent closed experiments. In 2016, CEEF was renamed to EEF (Ecology Experiment Facilities) to carry out isotopic chemical analysis experiments in the plant cabin. It will continue to study material cycles in BLSS and provide reference for their mechanisms (Tako, 2018).

3.4. Earth-based BLSS research: Europe

In 1989, funded by European Space Agency (ESA), Belgium, Spain, and Canada, the Micro-Ecological Life Support System Alternative (MELiSSA) was officially launched (Fulget et al., 1999). Managed by the ESA Technical Center, the research was carried out by multiple universities, research institutes and companies in close collaboration with research institutes of Germany, Ireland, the Netherlands and Russia. In the initial design, the MELiSSA system consisted of several interconnected compartments, including: the I–waste decomposition compartment, mainly composed of a thermophilic and anaerobic bacteria reactor; II–photosynthetic heterotrophic food production compartment, mainly composed of a photoheterotrophic bacteria reactor; III–nitrification compartment, mainly composed of nitrite bacteria and nitrifying bacteria reactor; IV–photosynthetic food production and atmospheric regeneration compartment, composed of a microalgae reactor (IVa) and higher plants (IVb); V–the crew compartment (Walker and Granjou, 2017). These compartments and equipment needed for the research are being designed and built by those various research institutions distributed in Europe, and unit experiments will be carried out.

MELiSSA plans to integrate these experimental facilities, and the integrated system will be set in University of Barcelona, Spain. A number of universities and research institutes have previously carried out researches in the fields of material circulation in the system (Farges et al., 2012) and efficient plant cultivation (Paradisoet al., 2014). The ultimate goal is to achieve the complete closure of water and air circulation in the system (Lasseur et al., 2010). Currently, MELiSSA project is proceeding cautiously and steadily, to build a complete crewed closed system in the future.

In addition, it is worth mentioning that the German Aerospace Center (DLR) studied the optimal structural design of a lunar greenhouse module (Vrakking et al., 2013). Considering the plant cultivation's requirements from greenhouse volume, mass, reliability, complexity and crew time (the working and waiting time needed before the greenhouse can be put into use), as well as challenges brought by lunar vehicle and complex environmental conditions, they used system engineering tools to assess designs in rigid and rigid-inflatable hybrid structures. The final optimal structure design they drew is a semi-cylindrical hybrid structure with rigid end-caps, 20 m in length and with a 3 m radius. Modeling result showed that the structure is compliant to requirements, with a dry and edible biomass yield of 22 kg/month.

3.5. Earth-based BLSS research: China

Since the mid-1990s, exploratory research on BLSS has been carried out in China. Many universities and research institutes have carried out a lot of work on international investigation of BLSS research and key unit technological research, and proposed the research direction and development plan of BLSS in China.

Since 2004, professor LIU Hong's research team at Beihang University has carried out systematic and in-depth research on BLSS. A lot of research on various key unit technologies has been done, such as higher plant cultivation (Hu et al., 2011), microalgae cultivation (Hu et al., 2008), treatment of plant inedible biomass (Yu et al., 2008), water cycling for utilization (Yang et al., 2008), animal selection (L. Li et al., 2013) and preparation of soil-like substrates (Kang et al., 2012). A small Earth-based "human-lettuce-microalgae-silkworm" experimental facility was established, and hereby a 279-day BLSS simulation experiment with partial human participation was carried out in China for the first time, with humans participating in the gas circulation by breathing the air of BLSS (M. Li et al., 2013). The experiments analyzed interrelationships among various species including plants, animals, microorganisms and microalgae (Tong et al., 2012; Tong et al., 2011, 2012), laying a solid foundation for subsequent studies. In October 2013, China's first Earth-based large-scale crewed comprehensive experimental BLSS system–"Lunar Place 1″ was built by the team, including a plant cabin and an integrated cabin, with a total area of 100 m^2, a total volume of 300 m^3 and a planting area of 69 m^2. In 2014, China's first crewed high-closure closed experiment was carried out in Lunar Palace 1, lasting for 105 days. Three crew members cultivated 21 types of grain and vegetable crops, cultivated yellow mealworm using biologically treated straw, and treated waste from all members inside the system, achieving a 100% regeneration of oxygen, a 100% regeneration of water, a 55% regeneration of food and an overall material closure of 97%, and thus successfully achieved the long-term stable operation of BLSS with such four biological units of "human–plant–animal–microbe" (Fu et al., 2016).

In 2016, Lunar Palace 1 completed the technological upgrade, with two plant cabins and one integrated cabin, a total area of 160 m^2, a volume of 500 m^3, and a planting area of 120 m^2. Hereby, the "Lunar Palace 365″ experiment started on May 10, 2017, and successfully completed on May 15, 2018, lasting for 370 days (Fu et al., 2021). In order to study the influence of crew shift changes and electromechanical failures on BLSS's stability under long-term operation condition, thus establishing the regulation technologies for long-term operation, two groups of crew with different metabolic levels were set up, forming three shift-change stages. The results showed that the BLSS of Lunar Palace 1 had good stability in the long-term operation process, and eliminated the influence of gas disturbance resulted from shift changes and electromechanical failures through self-feedback regulation, showing a strong robustness. The crew cultivated 35 kinds of plants, including grain, vegetable crops and berries. The plant production met 100% of the crew's need of plant food. Purification efficiency on sanitary wastewater reached the standard of plant irrigation, and urine and solid waste were also recycled. The experiment achieved a 100% regeneration of oxygen and water, an 83% regeneration of food and an overall material closure of 98.2% under the load of 4 crew members.

In addition to Lunar Palace 1 team, other several research institutes and universities in China also conducted research on BLSS. After years of research, the China Astronaut Research and Training Center established a small BLSS system in 2011, and completed a short-term crewed closed experiment with two people for 30 days on December 1, 2012. Later, they collaborated with SPACEnter Space Science and Technology Institute in the city of Shenzhen to build a large BLSS Earth-based experimental facility, and started the "Space 180″ experiment on June 17, 2016, which completed on December 14 of the same year, involving four crew members, lasting 180 days. The "Space 180″ experiment achieved a 100% regeneration of oxygen, a 99% regeneration of water and a 70% regeneration of food, and got research achievements in medicine (Yuan et al., 2019) and material circulation (Li et al., 2018). In addition, Fujian Academy of Agricultural Sciences adopted aquatic ecosystem and carried out a short-term experiment by using a biological loop of "human - *Azolla* - fish". The 50.4 m^2 aquaculture system basically achieved the oxygen requirement of two adults and 200 kg of fish (Chen et al., 2012).

4. Prospect of spatial BLSS experimental research

The ultimate goal of BLSS development is to realize long-term autonomous human survival in extraterrestrial space. The completed crewed closed experiments of BLSS so far are all Earth-based experiments. However, there are significant differences between the Earth's environment and extraterrestrial environment, such as in gravity, magnetic field, radiation and other aspects. Therefore, to truly apply extraterrestrial BLSS, experiments of terrestrial ecosystems must be carried out in extraterrestrial environmental conditions. However, this is still blank at present, and will thus become a major focus. Long-term operation experiments of small-scale terrestrial ecosystem must be carried out on space stations, the Moon, Mars and other extraterrestrial conditions to compare experimental data between extraterrestrial and the Earth's environment, through which calibration parameters and models will be obtained to correct the parameters and models obtained from large-scale Earth-based crewed experimental systems, to ultimately establish BLSS technologies that can be applied to space.

The Moon, as the only and near satellite of the Earth, is the most suitable extraterrestrial body for BLSS experiment. However, it is tidally locked by Earth, making most nights on its surface as nearly 14 Earth days long. Such long lunar nights will put forward extremely strict requirements on energy supply, thermal environment control and other functions of facilities operating on lunar surface. For a small BLSS operating on the lunar surface, stable energy supply is the primary need, leaving most areas on lunar surface not meeting this need.

However, due to the extremely small tilt of the Moon's rotation and the rugged terrain in lunar polar region, there will be continuous illumination areas (Li et al., 2008) near lunar north and south poles, especially the edge of the impact crater, where long-term and relatively stable light energy supply could exist. Such area will be exposed to sunlight at least 75% of the time in a year, and in some well-located areas, sunlight exposure duration in a year may even exceed 90% (Gläser et al., 2017). Correspondingly, there are permanent shadow areas at the bottom of some impact craters. These shadow areas cannot receive sunlight and are in a permanent cold and dark state. Theoretically, there may be water ice (Bussey et al., 2003), which has been supported by evidence (Feldman et al., 1998; Sefton-Nash et al., 2019).

Therefore, according to environmental conditions of lunar polar region, carrying out operation experiment of uncrewed closed ecosystem in this region has these feasibilities:

First, the favorable condition of long sunlight duration near the polar region basically enables "from seed to seed" (full growth cycle) plant cultivation in the system on lunar surface. In other areas of lunar surface, the closed ecosystem operation experiment would inevitably face a severe test of multiple lunar nights during the growth cycle of plants. If the location and timing of landing in lunar polar region are ideal, it is even possible to use seeds produced by the first generation of plants to cultivate the second and the third generation of plants *in situ*, to further study the effect of lunar environment on plant genetic traits.

Second, the sunlight condition of long duration and small solar angle in lunar polar region will avoid the unfavorable temperature condition of extreme fluctuation between high and low temperatures at other parts of lunar surface, with the temperature remained in a relatively small range, making the thermal control requirements relatively low. This not only indirectly improves the reliability of the system, but also further reduces payload mass and thus manufacturing and launch costs.

Third, in the long run, to build a large-scale, closed ecosystem on lunar surface, a large amount of substances necessary to support the operation of the system, such as water and plant cultivation substrates, are needed. If the resources of such substances could be mined and utilized on lunar surface *in situ*, the construction cost of the lunar base will be further reduced. Therefore, the closed ecosystem experiments carried out in the polar region should also study the utilization of lunar soil, water ice and other resources *in situ*.

Although the polar region theoretically has the basic conditions for closed ecosystem experiment, there are still a series of problems to be solved, such as the too-small solar angle in the polar region, the need for convergence of external horizontal illumination, and automatic control of the system. All these problems need to be verified in subsequent lunar reconnaissance activities and Earth-based simulation experiments, and corresponding solutions should be proposed, which will provide more theoretical and technological basis for subsequent development of the ultimately applied extraterrestrial BLSS.

5. Development path and evaluation indices of extraterrestrial BLSS

With the improvement and maturity of BLSS technologies, BLSS will enter extraterrestrial system construction in a real sense in the future. The development path of extraterrestrial BLSS can be summarized as a "three-stage strategy" (Fig. 4):

First stage. At the early stage of extraterrestrial BLSS construction, the plant cultivation in the system will mainly adopt hydroponics, and some *in situ* resources such as lunar soil and Martian soil will be processed together with the wastes in the system to prepare soil-like substrates, as a supplementary method of plant cultivation. The physical and chemical methods in the system can serve as the redundancy of the biological regeneration system and gas balance control, treating the waste together with the biological regeneration system. At this point, the system will provide the crew with all plant food needed, with oxygen and water fully regenerated.

Second stage. With the expansion of extraterrestrial BLSS, large amounts of soil-like substrates prepared from *in situ* resources will be

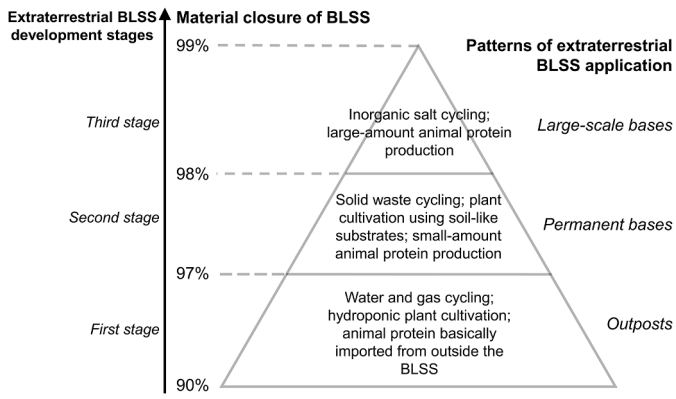

Fig. 4. Development path of extraterrestrial BLSS.

utilized, which will gradually become the major pattern for plant cultivation. Part of the plant's inedible biomass will be used to grow insects such as mealworms. At this point, the system will provide all plant food and part of animal protein for the crew, with oxygen and water fully regenerated.

Third stage. When the scale of BLSS is further expanded, inorganic salts in BLSS will be cycled. Organics will be purified from solid waste of crew members to produce salt water, which will be used to construct artificial ocean for fish cultivation, and the salt will be recovered by natural salt extraction. On this stage, BLSS will meet the needs of all plant and animal food for astronauts of long-term residence in the base, achieve a 100% biological regeneration of water, oxygen and other resources, with physical and chemical technologies only serving as the redundancy for gas balance regulation of biological regeneration; most of the waste will be treated biologically, while some recalcitrant waste hard to be treated biologically will be treated by physical and chemical technologies.

In addition, relevant standards and technological specifications of BLSS also need to be established and promoted simultaneously. To evaluate the development and technological level of BLSS, the following evaluation indices should be considered:

(1) The regeneration rate (%) of oxygen, water and food (including plant food and animal food) in BLSS; the material closure of BLSS (between 0 and 100%, the closer to 100%, the higher the autonomy);
(2) Stability and reliability of the operation of BLSS;
(3) Energy consumption and space required for life support per person (kWh•m^3/person);
(4) Weight of equipment required for life support per person (kg/person);
(5) Labor intensity required for life support per person (hour/person).

Specifically, gas balance is key to the stability of BLSS. Summarizing all BLSS research of more than 60 years around the world, the "three key conditions of BLSS gas balance" below should be complied for maintaining BLSS gas balance. These three key conditions will guide the construction of BLSS not only of Earth-based experiments, but also in future extraterrestrial bases.

First key condition: The food should be from biosynthesis in BLSS, and the oxygen produced should meet the crew's breathing need. *Second key condition*: Organic matter input from outside is tolerable, but it should be balanced by organic matter output of the same respiration quotient synthesized inside. *Third key condition:* The gas balance of BLSS should not be exclusively regulated by biological self-organization, and should be regulated by human intervention.

6. Overall prospect of BLSS development

To further improve and develop BLSS technology, new methods and thinking should be introduced to BLSS design, such as integrating and utilizing the latest technological progress in other fields. For example, BLSS can use flexible cabin technology to expand available internal space to provide larger biological cultivation area (Elsabbagh, 2015); apply growth promoting nanoparticles carrying activator protein (Gao et al., 2018), plant probiotics (Fu et al., 2017) and plant breeding technologies etc. to plant cultivation to achieve shorter growth cycle and higher yield; apply transgenic technology to microorganisms for higher waste disposal efficiency; further research on material resource cycle and introduction of more species into BLSS can not only improve the food closure, but also enrich the diet of the crew to further ensure their health (Hao et al., 2018); provide psychological care for the crew in such claustrophobic environments by means of plant horticulture therapy, photoperiodic regulation and appropriate psychological counseling (Li et al., 2020); rationally utilize the various *in situ* resources of target extraterrestrial bodies to reduce the requirement for transportation of life support substances (Gilrain et al., 1999; Kozyrovska et al., 2006); the internal environment of BLSS should be cleaned and disinfected regularly, and microbial prevention and control should be carried out to prevent cross contamination between the external environment and the cabin environment (Li et al., 2019; Yang et al., 2020), etc. In short, future development of BLSS will be comprehensively planned with the reality of lunar/Martian bases, the transportation capacity, various biological growth cycles, running environment of equipment, the crew's physical and psychological changes, and *in situ* resource utilization. BLSS will also be developed to directions such as lightweight, automated, real-time, and modular (Nelson et al., 2008).

According to historical experience, the development of space science and technology has a considerable driving effect on other research fields and the economy. Theoretical and technological achievements of BLSS can be applied not only to space, but also to life support needs in underwater and underground closed cabins, desert research stations, frontier posts and other areas with fragile ecosystem services and inconvenient material transportation, so as to minimize material transportation cost and the disturbance to the local environment. Derivative technologies of BLSS, such as efficient plant cultivation technology, plant straw and other solid waste fermentation technology can also be transferred to civilian fields to further improve agricultural efficiency and reduce environmental pollution, with broad prospects of development. The application of BLSS in civilian fields will also accumulate experience for future space applications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Corrigendum to 'Review of research into bioregenerative life support system(s) which can support humans living in space' [life sciences in space research 31 (2021) 113–120]

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The authors regret that the "three key conditions" in the final paragraph of Section 5 (page 6 of the article/ page 118 of the volume) is not sufficiently accurate. The revised corrected paragraph in full can be found below:

First key condition: If all food comes from BLSS's internal biosynthesis, the oxygen produced is sufficient for the crew's breathing need. *Second key condition:* If there is organic matter input from outside, it must be balanced by organic matter output of the same respiration quotient synthesized inside. *Third key condition:* The gas balance of BLSS cannot be achieved by biological self-organization and calls for human intervention.

The authors would like to apologise for any inconvenience caused.

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; BLSS, Bioregenerative Life Support System.

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